FEAT-ZD-48600 · DS-03 Database changes · Completed database changes

Data Science agent · item task.db_changes · source hobasa-data-science-agent/outputs/FEAT-ZD-48600/task.db_changes.md · approved by Developer sim.developer

task.db_changes: FEAT-ZD-48600 (Data Science lane, ap_forecast)

DS-03 Database changes (attempt 1). Workbook action: check the required database changes are completed.

Result: the Data Science lane needs no database change, so nothing is outstanding. The step outcome ("Required database changes are completed") is met because there are no changes to make.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	DS-03 Database changes (attempt 1)	record:brief
Lane / repository	Data Science / ap_forecast	feat.tasks (section 1, TSK-03, TSK-04)
Tasks in scope	TSK-03 (STORY-03, `risk.vendor_risk`), TSK-04 (STORY-04, `build_explanation_request(..., risk=None)`)	feat.tasks (sections 1, 3)
Inputs	feat.tasks (b0fd4d2), feat.design (722a98f)	record:brief

1. What the design says about storage

Check	Finding	Source
Data boundary of the feature	"Storage, database, schema migration" is listed as **out of this design**. The in-scope data is one new record key `paid_date`, one new reject reason `'paid_date'`, an in-memory delay history, risk map and overdue rows, and the CT-1 JSON	feat.design (A1, Data row)
Technology	No new repository, service, runtime, library or storage; only the Python 3.9 standard library and Node.js 22 CommonJS (DD-1)	feat.design (A7, DD-1); feat.tasks (intro)
Purity	Every new function is pure and in-memory: no I/O, no network, no mutation of inputs (NF-10)	feat.design (A5, NF-10)
Platform trigger	"any storage or network call from the libraries" would reopen DES-01; the design triggers none	feat.design (A8)
ap_forecast modules	New `src/ap_forecast/risk.py`; additive `risk=None` in `explain.py`; `__init__.py` export and version step; `predict.py` and `models.json` unchanged	feat.design (B2.1)
ap_forecast data structures	`history` (iterable of `int`), risk level (`str`), explanation request (`dict`), all held in memory	feat.design (B2.3)
TSK-03 constraints	No LLM, no network, no `models.json` read	feat.tasks (TSK-03 Goal)
TSK-04 constraints	Provider, model and `max_tokens` still come from `config or load_config()`; `models.json` must not change	feat.tasks (TSK-04 Must not change); feat.design (B2.2)
Package version today	ap_forecast `0.9.0`; the version step is owned by TSK-03. This is a package version, not a schema version	feat.design (B2 heading); feat.tasks (section 2)

2. What the repository holds today

Check	Finding	Source
Database or migrations	ap_forecast has no database; no migrations exist or are required	tool:repo-access
Tracked source files	`src/ap_forecast/__init__.py`, `explain.py`, `predict.py`, `models.json`; no migration folder, schema or SQL file	file:ap_forecast/src/ap_forecast (directory listing)
Database usage in code	A case-insensitive search of the repository for `sqlite`, `sqlalchemy`, `psycopg`, `migrat`, `database` and `CREATE TABLE` found no match	file:ap_forecast (search)
`predict_delay_days`	Pure computation over the given history (rounded mean, negatives count as 0); no storage	file:ap_forecast/src/ap_forecast/predict.py
`build_explanation_request`	Reads only the packaged `models.json` configuration file through `load_config()`; it does not touch a database	file:ap_forecast/src/ap_forecast/explain.py

3. Required database changes

#	Change	Status
—	None required by TSK-03 or TSK-04	Complete (nothing to do)

The Data Science lane needs nothing from DBAs or infrastructure. This step adds no migration, schema, seed data or connection secret (feat.design A1, DD-1; tool:repo-access).

4. Re-entry triggers

Re-run this check if any of the following appears. Each would bring storage into ap_forecast and also reopen DES-01 (feat.design A8):

- `vendor_risk` results or delay histories are to be persisted instead of computed in memory.
- The production wiring of the stages (open question DQ-1 in feat.design D3) introduces a store or service.
- A provider or model change (PQ-5, TQ-8) moves configuration out of `models.json` into a store.

5. Open questions

None new. The carried design questions (DQ-1 to DQ-5, TQ-1 to TQ-11) are unchanged, and none of them concerns a database for this lane.

[Tool result] [ap_forecast] No database in this repository's architecture; no migrations required